3C: Bundling

Industrial Organization

Renmin University of China

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Overview

1. Introduction
2. Pure Bundling
3. Nested Bundling
4. Partition Bundling
5. Conclusion

1 Introduction

2 Pure Bundling

3 Nested Bundling

4 Partition Bundling

5 Conclusion

Bundling

- Bundling means packaging multiple products or services together for sale.
  - A common practice in the real world:
    1. Microsoft Office 365: Word, PowerPoint, Excel...
    2. Netease Music: Music Access, VIP, SVIP.
    3. A stationer's: ruler set, English textbook bundle...
- Q: What makes bundling a profitable selling strategy? When is bundling the optimal selling strategy?

Outlines

Today's course covers three types of bundling, each with a seminal paper.

- *When Is Pure Bundling Optimal?* (Haghpanah and Hartline, *RES*, 2021)
- *Nested Bundling* (Yang, *AER*, 2025)
- *Robustly Optimal Mechanisms for Selling Multiple Goods* (Che and Zhong, *RES*, 2025)
  - Partition bundling

Setup

- A monopolistic seller of products $\{1, \ldots, n\}$
  - Production costs are 0.
  - $B = 2^{\{1,2,\cdots,n\}}$ denotes the set of all bundles.
- A mass 1 of buyers who need at most 1 unit of each product
  - Buyer's utility $v(b, t) - p$ from consuming b and paying p
    - Buyer's privately known type $t \in T = [\underline{t}, \bar{t}]$, with distribution F
    - Value $v(b, t) \geq 0$ for each bundle $b \subseteq \{1, 2, \ldots, n\}$, $v(\varnothing, t) = 0$
    - Non-additive values vs. additive values $v(b, t) = \sum_{i \in b} v(\{i\}, t)$
  - Denote the expected value of a randomized bundle $a \in \Delta(B)$ for type t by $v(a, t) = \mathbb{E}_{b \sim a}[v(b, t)]$.

Selling Mechanisms

- A direct mechanism is a pair of a bundle assignment rule $q : T \to \Delta(B)$ and a payment rule $t : T \to \mathbb{R}$.
  - Incentive compatibility and individual rationality:

$$v(a(t), t) - p(t) \geq v(a(t), t') - p(t'), \quad \forall t, t' \in T. \tag{IC}$$

$$v(a(t), t) - p(t) \geq 0, \quad \forall t \in T. \tag{IR}$$

  A feasible (IC and IR) mechanism is optimal if it maximizes $\mathbb{E}[p(t)]$.

- $\mathcal{M}$ denotes the set of feasible mechanisms.

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Setup

- Q: To maximize expected profit, when should Seller use pure bundling, i.e., offer only the grand bundle $\bar{b} = \{1, 2, \cdots, n\}$?
  - It depends on how Buyer's relative values, the ratio $r(b, t) = v(b, t)/v(\bar{b}, t)$ for each bundle b, change with Buyer's value for the grand bundle $v(\bar{b}, t)$.
  - The profile of relative values for a type t is a function $r(\cdot, t) \in \mathbb{R}^B$.
    - For a set S, $\mathbb{R}^S$ denotes the set of functions from S to $\mathbb{R}$.

When is Pure Bundling Optimal? (Two Types)

Consider the binary case: $T = \{t_L, t_H\}$.

Corollary 1

Pure bundling is optimal if the low type t_L has lower relative values than the high type t_H does, that is, $r(b, t_L) \leq r(b, t_H)$ for all bundles b.

- For example, assume that $v(b, t_H) - 1 \leq v(b, t_L) \leq v(b, t_H)$ for each bundle b, $v(\bar{b}, t_L) = 1$, and $v(\bar{b}, t_H) = 2$.
- The highest revenue among pure bundling mechanisms is $\max\{1, 2q_H\}$.
  - $q_H = \mathbf{Pr}[t = t_H]$ denotes the probability of the high type.
    - If the price of $\bar{b}$ is 1, then both types buy $\bar{b}$, and the revenue is 1.
    - If the price of $\bar{b}$ is 2, then only the high type buys $\bar{b}$, and the revenue is $2q_H$.
    - Any other price results in revenue less than 1 or $2q_H$.

When is Pure Bundling Optimal? (Two Types)

- Consider a "mixed bundling" mechanism that offers some bundle b at price $v(b, t_L)$ and $\bar{b}$ at price $2 - \big(v(b, t_H) - v(b, t_L)\big)$[1].
  - Type t_L chooses bundle b, and type t_H chooses $\bar{b}$.
- The revenue is $v(b, t_L) + q_H\big(2 - v(b, t_H)\big)$.
  - This is at most $\max\{1, 2q_H\}$ if $r(b, t_L) \leq r(b, t_H)$, that is, $v(b, t_L) \leq \frac{1}{2}v(b, t_H)$.
    - $q_H \geq \frac{1}{2}$: The revenue of mixed bundling is

$$\begin{aligned} v(b, t_L) + q_H\big(2 - v(b, t_H)\big) &= v(b, t_L) + 2q_H - q_H v(b, t_H) \\ &\leq v(b, t_L) + 2q_H - v(b, t_L) = 2q_H. \end{aligned}$$

    - $q_H \leq \frac{1}{2}$: The revenue of mixed bundling is

$$\begin{aligned} v(b, t_L) + q_H\big(2 - v(b, t_H)\big) &\leq v(b, t_L) + \frac{1}{2}\big(2 - v(b, t_H)\big) \\ &\leq v(b, t_L) + 1 - v(b, t_L) = 1. \end{aligned}$$

[1] The price of $\bar{b}$ is set to maximize revenue while satisfying IC.

When is Pure Bundling Optimal?

- A set $U \subseteq \mathbb{R}^S$ is an upper set if $x' \leq x$ and $x' \in U$ imply that $x \in U$.

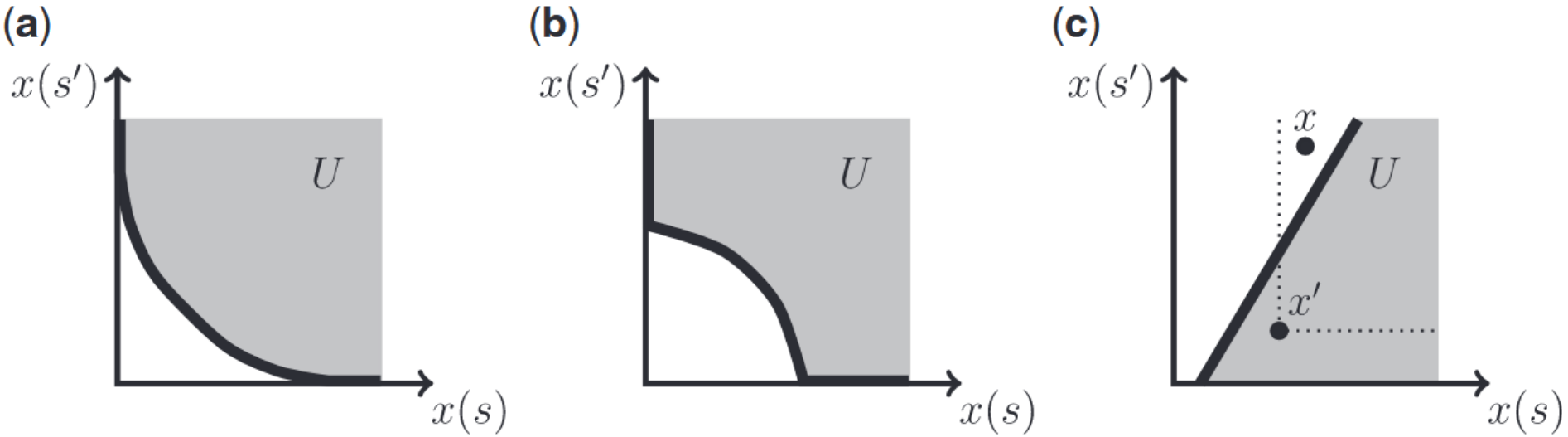

FIGURE 1

Sets $U \subseteq \mathbb{R}^{\{s,s'\}}$ in (a) and (b) are upper sets. The set $U \subseteq \mathbb{R}^{\{s,s'\}}$ in (c) is not an upper set.

When is Pure Bundling Optimal?

Theorem 1

Pure bundling is optimal if the profile of relative values is stochastically non-decreasing in positive values of the grand bundle; i.e.,

$$\mathbf{Pr}\left[r(\cdot, t) \in U \mid v(\bar{b}, t) = \hat{v}\right]$$

is non-decreasing in $\hat{v} > 0$ *for all upper sets* $U \subseteq \mathbb{R}^B$.

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Technical Assumptions

- Nested bundling: Larger bundles include all the goods of the smaller ones.
- Continuous density $f > 0$
- Assume efficiency for type $\bar{t}$ to buy $\bar{b}$, i.e., $\arg\max_b \{v(b, \bar{t})\} = \bar{b}$.
- $v(b, t)$ is (i) nondecreasing in b in the set-inclusion order; (ii) continuously differentiable in t; (iii) strictly increasing in t whenever $v(b, t) > 0$.

Partial Order

- The sold-alone quantity $Q(b)$ is defined as the unique quantity that maximizes Seller's revenue when he only offers b.
- Define the following partial order on the set of bundles:

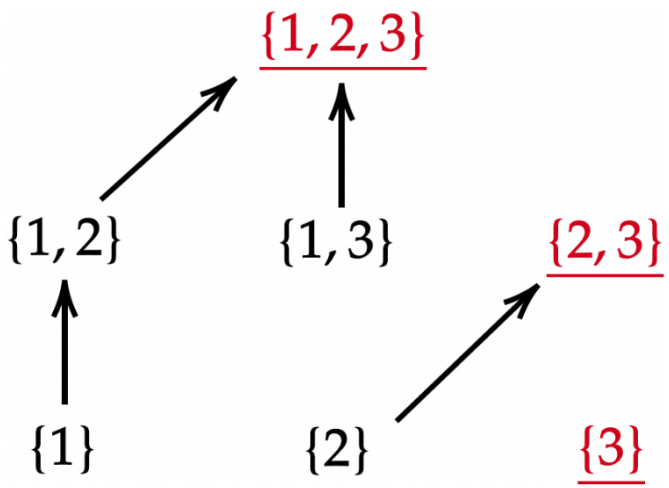
$$b_1 \preceq b_2 \iff b_1 \subseteq b_2 \text{ and } Q(b_1) \leq Q(b_2).$$

A bundle b is dominated if there exists $b' \neq b$ such that $b \preceq b'$ and undominated otherwise.

Nesting Condition

- The nesting condition holds if the undominated bundles can be totally ordered by set inclusion: for any two bundles b and b',

$$\text{both } b \text{ and } b' \text{ are undominated} \;\Rightarrow\; \text{either } b \subseteq b' \text{ or } b' \subseteq b. \qquad \text{(Nesting Condition)}$$



Intuition

- Assume that consumer types are one-dimensional. [2]
  - A higher-type consumer has higher incremental values for larger bundles.

Assumption 1 (Incremental Monotonicity)

For any two nested bundles $b_1 \subset b_2$, the incremental value $v(b_2, t) - v(b_1, t)$ is strictly increasing in t whenever it is strictly positive.

- A key feature of the one-dimensional type space: We can compute the total revenue from any feasible allocation, induced by some menu, from the sold-alone marginal revenue curve.
  - $P(b, q)$ denotes the demand curve for bundle b when it is sold alone.
  - $MR(b, q)$ denotes the corresponding marginal revenue curve for bundle b.

[2] However, the problem of bundling is multidimensional, because screening can be done using multiple instruments.

Intuition

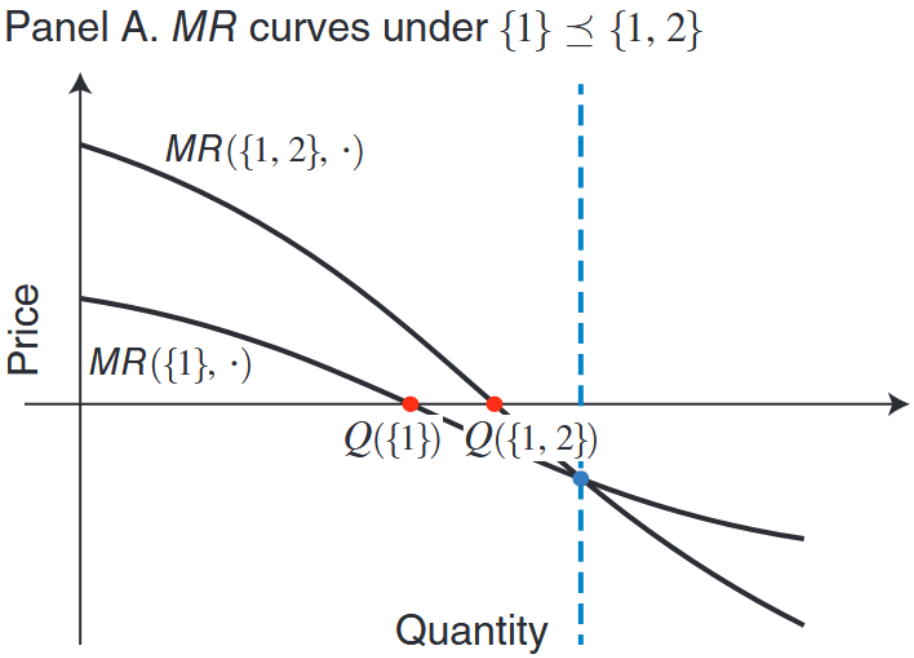
- We can arrange consumers along a single quantity axis, with consumers positioned toward the right end having lower values for all the bundles.
  - Given a menu, they optimally choose their options.
    - Allocation rule $b(q)$ describes the bundle choice for the consumer at quantity q.
    - $CS(q)$ denotes the surplus of the consumer at quantity q, $CS(1) = 0$.
- The total revenue is thus

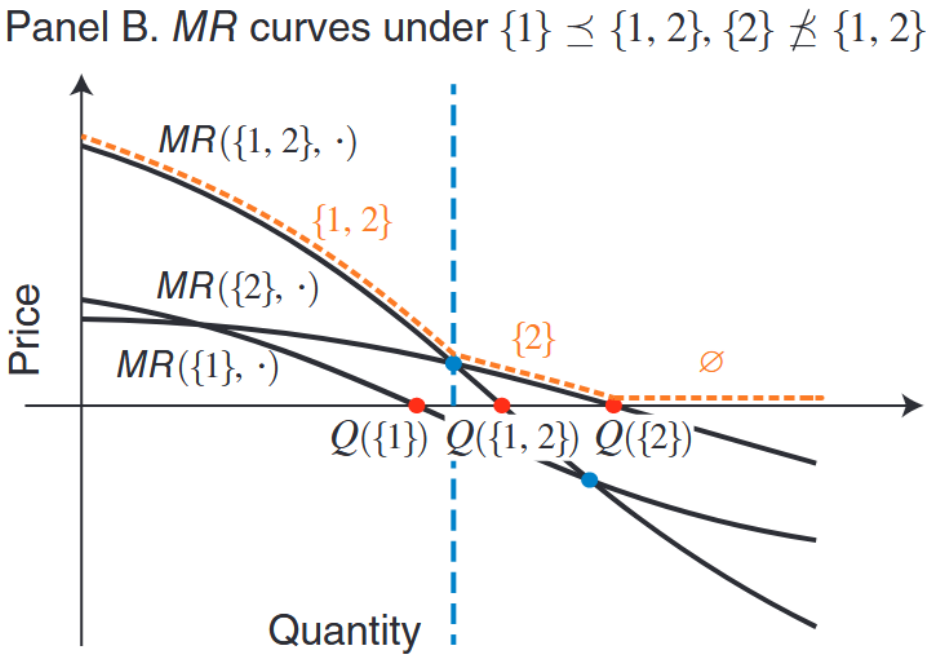
$$\begin{aligned}
\text{TR} &= \int_0^1 P(b(q), q)\mathrm{d}q - \int_0^1 CS(q)\mathrm{d}q && \text{(Value} - \text{Consumer Surplus)} \\
&= \int_0^1 P(b(q), q)\mathrm{d}q + \int_0^1 q \cdot CS'(q)\mathrm{d}q && \text{(Integration by Parts)} \\
&= \int_0^1 P(b(q), q)\mathrm{d}q + \int_0^1 q \cdot P'_q(b_q, q)\mathrm{d}q && \text{(Envelop Theorem)} \\
&= \int_0^1 MR(b(q), q)\mathrm{d}q.
\end{aligned}$$

Intuition

- Suppose we have two items $\{1, 2\}$ and costs are 0.
  - Suppose their sold-alone quantities satisfy $Q(\{1\}) < Q(\{1, 2\}) < Q(\{2\})$.
    - $\{1\} \preceq \{1, 2\}$, but $\{2\}$ is not dominated by $\{1, 2\}$.
    - The two undominated bundles are nested.
- Suppose the revenue function from selling any incremental bundle (i.e., the option to upgrade from a smaller bundle to a larger one) is strictly quasi-concave.
  - Geometric interpretation:
    - The MR curves cross zero once from above.
    - The MR curve of a larger bundle also crosses the MR curve of a smaller bundle at most once from above.

Intuition


Panel A. MR curves under $\{1\} \preceq \{1, 2\}$
$MR(\{1, 2\}, \cdot)$
$MR(\{1\}, \cdot)$
Price
$Q(\{1\})$
$Q(\{1, 2\})$
Quantity


Panel B. MR curves under $\{1\} \preceq \{1, 2\}, \{2\} \not\preceq \{1, 2\}$
$MR(\{1, 2\}, \cdot)$
$\{1, 2\}$
$MR(\{2\}, \cdot)$
$\{2\}$
$\varnothing$
$MR(\{1\}, \cdot)$
Price
$Q(\{1\})$
$Q(\{1, 2\})$
$Q(\{2\})$
Quantity

Intuition

- Three observations:
  1. Because $Q(\{1\}) < Q(\{1,2\})$, the two quantities must be in the region where the MR of upgrading consumers from $\{1\}$ to $\{1,2\}$ is *positive*.
     - If it is profitable to sell a consumer $\{1\}$, which happens before $Q(\{1\})$, it is even more profitable to *upgrade* the consumer to $\{1,2\}$.
  2. Because $Q(\{2\}) > Q(\{1,2\})$, the two quantities must be in the region where the MR of upgrading consumers from $\{2\}$ to $\{1,2\}$ is *negative*.
     - After a certain quantity threshold, there always exists a region in which it is more profitable to *downgrade* the consumers from $\{1,2\}$ to $\{2\}$.
  3. With the upgrading and downgrading, we can attain the upper envelope of the MR curves by allocating bundles to consumers in a *monotone* fashion.
     - A higher-type consumer receives a larger bundle in the set-inclusion order.
     - We implement this using upgrading as follows: (1) Set the price of $\{2\}$ to be its sold-alone monopoly price; (2) Set the price of upgrading from $\{2\}$ to $\{1,2\}$ to be such that the consumer located at the threshold is indifferent to upgrading.

Local Quasiconcavity

- Under assumption (A1), for any two nested bundles $b_1 \subset b_2$, the difference $\pi(b_2, q) - \pi(b_1, q)$ is the demand curve generated by the incremental values for b_2 given b_1.
  - $\pi(b_2, q) - \pi(b_1, q)$ is the profit function of a monopolist optimizing the quantity of the incremental bundle $b_2 \setminus b_1$, given the plan of selling b_1.

Assumption 2 (Local Quasiconcavity)

For any two nested bundles $b_1 \subset b_2$, the incremental profit $\pi(b_2, q) - \pi(b_1, q)$ is strictly quasiconcave in $q \in [0, \min\{Q(b_1), Q(b_2)\}]$.

- The interval $[0, \min\{Q(b_1), Q(b_2)\}]$ is exactly the region where both profit functions $\pi(b_1, q)$ and $\pi(b_2, q)$ are increasing.

When is Nested Bundling Optimal?

Theorem 2

Suppose that assumptions (A1) and (A2) hold. Then, under the nesting condition,

(i) Nested bundling is optimal.

(ii) A menu of undominated bundles is optimal.

(iii) Every optimal mechanism is equivalent to nested bundling.

- A sufficient and more testable condition for nesting is as follows:

$$\text{For all } b_1 \text{ and } b_2,\ Q(b_1 \cup b_2) \geq \min\{Q(b_1), Q(b_2)\}$$

(Union Quantity Condition)

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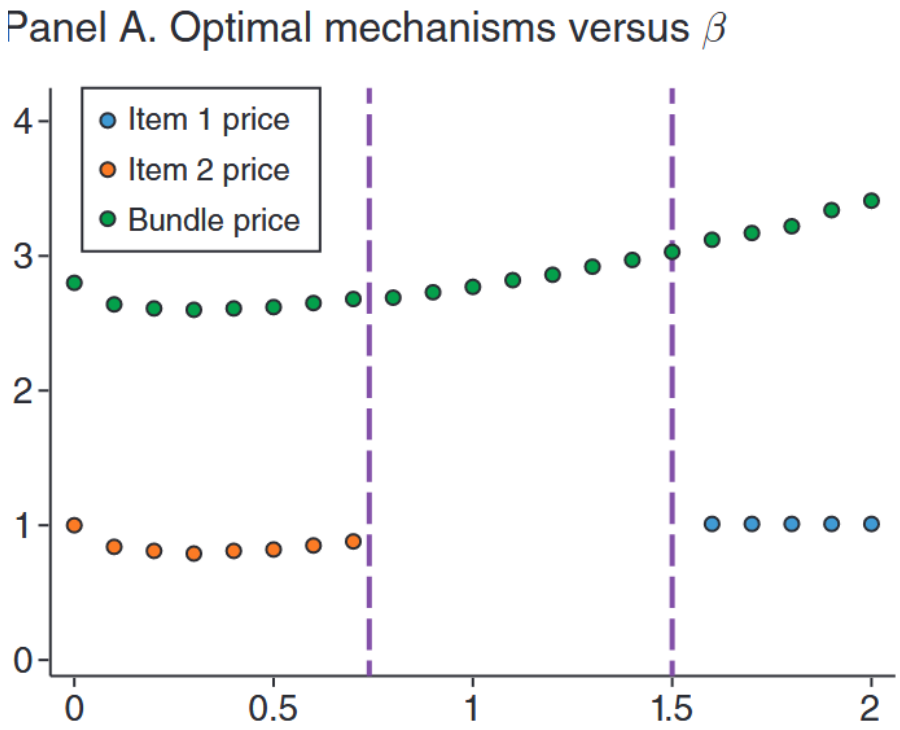
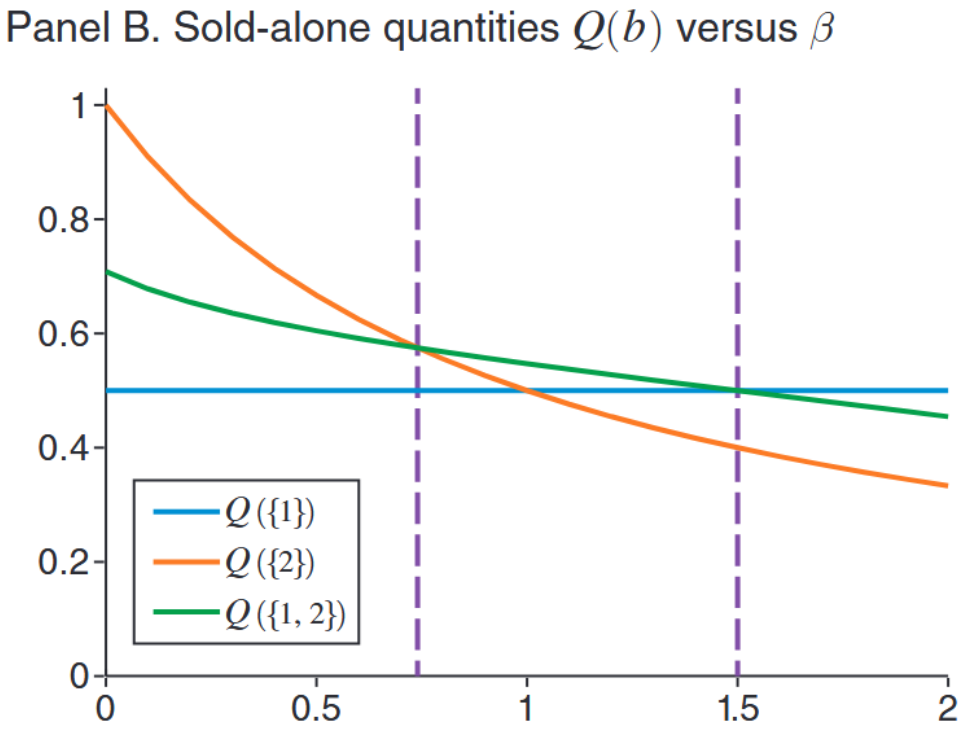
(Union Quantity Condition)

A Parametric Example

- Consider two items $\{1, 2\}$ and zero costs.
  - $v(\{1\}, t) = t$, $v(\{2\}, t) = t^\beta$, $v(\{1, 2\}, t) = t + t^\beta + \sqrt{t}$
  - $t \sim U[0, 2]$
  - Parameter $\beta \in [0, 2]$
- The nesting condition holds for every β.
- The optimal menu (of undominated bundles) is given by
  - $\{\{2\}, \{1, 2\}\}$ when $\beta \in [0, 0.74)$
  - $\{\{1, 2\}\}$ when $\beta \in [0.74, 1.5]$
  - $\{\{1\}, \{1, 2\}\}$ when $\beta \in (1.5, 2]$

A Parametric Example





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Ambiguity Set

- Additive values: Buyer's privately known values $\boldsymbol{v} := (v_1, \ldots, v_n)$.
- Seller simply knows that the distribution lies within some ambiguity set $\mathcal{F} \subset \Delta(\mathbb{R}_+^n)$ defined by a set of moment conditions.
- Moment conditions: Fix any joint distribution $F \in \Delta(\mathbb{R}_+^n)$. Assume that:
  - Seller has some knowledge about the means of item values.
    - $\mu_i(F) := \mathbb{E}_F[v_i]$ denotes the mean value of item i.
  - Seller has some knowledge about the dispersion of values of arbitrary subsets of items.
    - Let $\mathcal{K}$ be an arbitrary partition of the goods.
    - For each $K \in \mathcal{K}$, let

$$\sigma_K(F) := \mathbb{E}_F\left[\phi_K\left(\sum_{i \in K} v_i\right)\right],$$

be the dispersion of category K's value under F, where $\phi_K : \mathbb{R}_+ \to \mathbb{R}_+$ is a twice-differentiable convex function satisfying $\phi_K'' \geqslant \varepsilon$ for some $\varepsilon > 0$.

Ambiguity Set

- Therefore, Seller knows item value means and dispersion lie in some arbitrary nonempty convex and compact set $\Omega \subset \mathbb{R}_+^{n+|\mathcal{K}|}$.
- Seller faces an ambiguity set

$$\mathcal{F} := \{F \in \Delta(\mathbb{R}_+^n) : (\mu_i(F), \sigma_K(F))_{i \in N, K \in \mathcal{K}} \in \Omega\}. \tag{1}$$

Example. Suppose Seller has 3 goods $\{1, 2, 3\}$. She only knows the means $\mathbb{E}[v_1] = 0.5$, $\mathbb{E}[v_2] = \mathbb{E}[v_3] = 0.3$, and variances $\mathbb{V}[v_1] = \mathbb{V}[v_2 + v_3] = 0.1$ for two categories given by the partition $\mathcal{K} = \{\{1\}, \{2, 3\}\}$. The convex moment functions are $\phi_{\{1\}}(v) = \phi_{\{2,3\}}(v) = v^2$. Then the moment constraint set is $\Omega = \{(0.5, 0.3, 0.3, 0.35, 0.46)\}$.

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Robustness Solution Concept

- A feasible mechanism $M = (q, t) \in \mathcal{M}$ is a $\mathcal{K}$-bundled sales mechanism if, for each $K \in \mathcal{K}$, there exists a feasible mechanism $q_{\mathcal{K}} : \mathbb{R}_+ \to [0, 1]$ and $t_{\mathcal{K}} : \mathbb{R}_+ \to \mathbb{R}$ such that $t(\mathbf{v}) = \sum_{K \in \mathcal{K}} t_K \left(\sum_{j \in K} v_j \right)$ and $q_i(\mathbf{v}) q_{K(i)} \left(\sum_{j \in K} v_j \right)$.
- Seller's revenue from a mechanism $M \in \mathcal{M}$ given value distribution F is

$$R(M, F) = \int t(v) \mathrm{d}F(v).$$

Let

$$R^* = \sup_{M \in \mathcal{M}} \inf_{F \in \mathcal{F}} R(M, F) \tag{2}$$

We say the mechanism attaining R^* is robustly optimal.

Robustly Optimality of Partition Bundling

Theorem 3

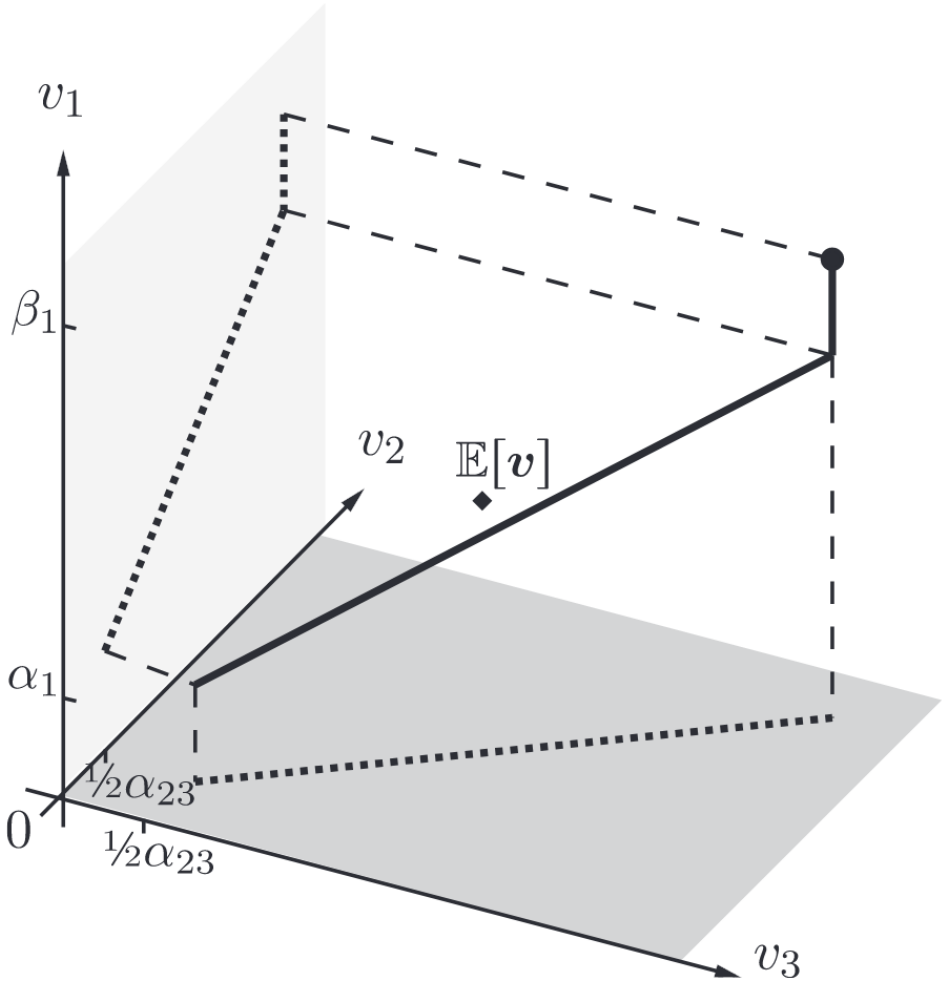
It is robustly optimal for the seller to use a $\mathcal{K}$-bundled sales mechanism.

Intuition. (Based on Example 1) Construct a $\mathcal{K}$-bundled sales mechanism $M^* \in \mathcal{M}$ and distribution $F^* \in \mathcal{F}$ such that they form mutual best responses.

Nature chooses $F^* \in \mathcal{F}$ that has a one-dimensional support: the support forms a ray starting from the origin and contains a vertical segment toward the end.

- Specifically, nature draws a one-dimensional random variable X distributed from $[1, \infty)$ according to CDF $1 - \frac{1}{x}$ (Pareto distribution).
- The valuation of each good $i \in \{1, 2, 3\}$ is $v_i = \min\{\alpha_i X, \beta_i\}$
  - (α_i, β_i)'s are chosen uniquely to satisfy the moment conditions.

Intuition for Theorem



Intuition for Theorem

Two properties notable about F^*:

1. The support is comonotonic within each category.
   - This property incentivizes Seller to bundle each category of goods.
2. F^* is designed to suppress the seller's revenue as much as possible.
   - Since F^* is one-dimensional, the seller's mechanism design problem reduces to a standard one-dimensional screening problem.
   - The virtual value of each item is $J_i(x) = V_i(x) - V_i'(x)\dfrac{1 - H(X)}{h(x)}$.
     - The "flat"virtual value function means that all mechanisms that allocate each item i to buyer type with valuation β_i are optimal.
     - Since any $\mathcal{K}$-bundled sales mechanism that sets bundle $\{1\}$'s price in $[\alpha_1, \beta_1)$ and bundle $\{2, 3\}$'s price in $[\alpha_{23}, \beta_{23}) = [\alpha_2 + \alpha_3, \beta_2 + \beta_3)$ satisfy this property, M^* constructed below is optimal given F^*.

Intuition for Theorem

The optimal $\mathcal{K}$-bundling mechanism M^* sells the two categories $\{1\}$ and $\{2,3\}$ as two separate bundles, and prices the respective bundles independently and randomly according to density functions:

$$g_1(p) = 2\lambda_1 \cdot \frac{\beta_1 - p}{p}$$

on support $[\alpha_1, \beta_1]$, and

$$g_{23}(p) = 2\lambda_{23} \cdot \frac{\beta_{23} - p}{p}$$

on support $[\alpha_{23}, \beta_{23}]$, and $\lambda = (\lambda_1, \lambda_{23})$ are scale factors chosen so that the densities integrate to ones.

Intuition for Theorem

Intuitively, the $\mathcal{K}$-bundling mechanism is designed to "hedge"against possible deviation by nature away from F^*.

- Recall what nature is capable of doing:
  1. Nature can freely shift values across items within the bundle $\{2, 3\}$, as the moment condition only constrains its total value.
  2. Nature can freely correlate the values of categories $\{1\}$ and $\{2, 3\}$.
  3. Nature can redistribute the value of each bundle—$\{1\}$ and $\{2, 3\}$—while maintaining the same dispersion measures.

Intuition for Theorem

- Our mechanism makes it unprofitable for nature to deviate along these three channels.
  1. The bundling of items 2 and 3 makes their valuations perfectly substitutable from the revenue standpoint; hence, nature has no incentive to redistribute values within the $\{2, 3\}$ bundle.
  2. The independent pricing of the two bundles means that nature never gains from manipulating the correlations of values across the bundles.
  3. The prices of the two bundles are randomized according to the particular hyperbolic forms of the densities precisely to eliminate any incentive by nature to redistribute value within each bundle.

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Conclusion

- Pure bundling
  - Seller only offers the grand bundle.
  - Optimal if the profile of relative values is stochastically non-decreasing in the values of the grand bundle
- Nested bundling
  - Larger bundles include all the products of the smaller ones.
  - Optimal when the nesting condition holds
- Partition bundling
  - Seller divides its products into disjoint bundles.
  - Robustly optimal under additive values